Supplementary Material of Hu et al., "Geo-knowledge-guided GPT models improve the extraction of location descriptions from disaster-related social media messages", *International Journal of Geographical Information Science*, 2023

Table S1. Complete prompt created based on the geo-knowledge of common forms of location descriptions.

This is a set of location description recognition problems.
The `Sentence` is a sentence containing location descriptions.
The goal is to infer which parts of the sentence represent location descriptions and the categories of the location descriptions. Split different location descriptions with `;`.
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Sentence: Papa stranded in home. Water rising above waist. HELP 812 Wood Ln, 77828 #houstonflood
Q: Which parts of this sentence represent location descriptions?
A: C1: 812 Wood Ln, 77828
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Sentence: Anyone doing high water rescues in the Pasadena/Deer Park area? My daughter has been stranded in a parking lot all night
Q: Which parts of this sentence represent location descriptions?
A: C10: Pasadena/Deer Park
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Sentence: Allen Parkway, Memorial, Waugh overpass, Spotts park and Buffalo Bayou park completely under water
Q: Which parts of this sentence represent location descriptions?
A: C2: Allen Parkway; C2: Memorial; C2: Waugh overpass; C7: Spotts park; C7: Buffalo Bayou park
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Sentence: Streets Flooded: Almeda Genoa Rd. from Windmill Lakes Blvd. to Rowlett Rd. HurricaneHarvey Houston
Q: Which parts of this sentence represent location descriptions?
A: C11: Almeda Genoa Rd. from Windmill Lakes Blvd. to Rowlett Rd.; C9: Houston
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Sentence: Cleaning supply drive is underway. 9-11 am today at Preston Hollow Presbyterian Church
Q: Which parts of this sentence represent location descriptions?
A: C8: Preston Hollow Presbyterian Church
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Sentence: #Harvey LIVE from San Antonio, TX. Fatal car accident at Ingram Rd., Strong winds.
Q: Which parts of this sentence represent location descriptions?
A: C9: San Antonio; C9: TX; C2: Ingram Rd.
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Sentence: 9:00AM update video from Hogan St over White Oak Bayou, I-10, I-45: water down about 4' since last night.
Q: Which parts of this sentence represent location descriptions?
A: C5: Hogan St over White Oak Bayou; C3: I-10; C3: I-45
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Sentence: Left Corpus bout to be in San Angelo #HurricaneHarvey Y'all be safe Avoided highway 37 Took the back road
Q: Which parts of this sentence represent location descriptions?
A: C9: Corpus; C9: San Angelo; C3: highway 37
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Sentence: Need trailers/trucks to move dogs from Park Location: Whites Park Pavillion off I-10 exit 61 Anahuac TX
Q: Which parts of this sentence represent location descriptions?
A: C7: Whites Park Pavillion; C3: I-10; C4: exit 61; C9: Anahuac; C9: TX
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Sentence: Townsend exit, Sorters road and Hamblen road is flooded coming from 59 southbound HurricaneHarvery Harvey2017
Q: Which parts of this sentence represent location descriptions?
A: C4: Townsend exit; C5: Sorters road and Hamblen road; C3: 59 southbound
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Sentence: #Harvey does anyone know about the flooding conditions around Cypress Ridge High School?! #HurricaneHarvey
Q: Which parts of this sentence represent location descriptions?
A: C8: Cypress Ridge High School
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Sentence: FYI to any of you in NW Houston/Lakewood Forest, Projections are showing Cypress Creek overflowing at Grant Rd
Q: Which parts of this sentence represent location descriptions?
A: C10: NW Houston/Lakewood Forest; C5: Cypress Creek overflowing at Grant Rd
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Sentence: #HurricaneHarvey family needs rescuing at 11800 Grant Rd. Apt. 1009 in Cypress, Texas 77429, 2 elderly, one is 90 not steady in her feet
Q: Which parts of this sentence represent location descriptions?
A: C1: 11800 Grant Rd. Apt. 1009 in Cypress, Texas 77429
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Sentence: Guys, this is I-45 @ Main Street in Houston. Crazy. #hurricane #harvey. . .
Q: Which parts of this sentence represent location descriptions?
A: C5: I-45 @ Main Street; C9: Houston
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Sentence: Frontage Rd at the river #hurricaneHarvey #hurricaneharvey @ San Jacinto River
Q: Which parts of this sentence represent location descriptions?
A: C2: Frontage Rd; C6: San Jacinto River
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Sentence: Pictures of downed trees and damaged apartment building on Airline Road in Corpus Christi.
Q: Which parts of this sentence represent location descriptions?
A: C2: Airline Road; C9: Corpus Christi
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Sentence: Buffalo Bayou holding steady at 10,000 cfs at the gage near Terry Hershey Park
Q: Which parts of this sentence represent location descriptions?
A: C6: Buffalo Bayou; C7: Terry Hershey Park
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Sentence: Major flooding at Clay Rd & Queenston in west Houston. Lots of rescues going on for ppl trapped...
Q: Which parts of this sentence represent location descriptions?
A: C5: Clay Rd & Queenston; C9: Houston

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Sentence: HELP! A pregnant lady is stuck in her car on I-45 between Cypress Hill & Huffmeister exits #harvey
Q: Which parts of this sentence represent location descriptions?
A: C11: I-45 between Cypress Hill & Huffmeister exits
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Sentence: If you need a place to escape #HurricaneHarvey, The Willie De Leon Civic Center: 300 E. Main St in Uvalde is open as a shelter
Q: Which parts of this sentence represent location descriptions?
A: C7: The Willie De Leon Civic Center; C1: 300 E. Main St in Uvalde
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Sentence: Houston's Buffalo Bayou Park - always among the first to flood. #Harvey
Q: Which parts of this sentence represent location descriptions?
A: C9: Houston; C7: Buffalo Bayou Park
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Sentence: #HurricaneHarvey INTENSE eye wall of category 4 Hurricane Harvey from Rockport, TX
Q: Which parts of this sentence represent location descriptions?
A: C9: Rockport; C9: TX
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Sentence: {TEXT}
Q: Which parts of this sentence represent location descriptions?
A:

Table S2. Complete prompt created based on a different set of 22 tweet examples using the same geo-knowledge.

This is a set of location description recognition problems.
The `Sentence` is a sentence containing location descriptions.
The goal is to infer which parts of the sentence represent location descriptions and the categories of the location descriptions. Split different location descriptions with `;`.
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Sentence: Pls rescue 12 day baby family at 7 Woodview St, Houston, 77124, flooding will reach roof soon…
Q: Which parts of this sentence represent location descriptions?
A: C1: 7 Woodview St, Houston, 77124
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Sentence: What's left of my front and back yard in the League City/Dickinson area.
Q: Which parts of this sentence represent location descriptions?
A: C10: League City/Dickinson
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Sentence: Thanks Home Depot in Plano on Custer Rd for your help with Flood Cleanup Kits!
Q: Which parts of this sentence represent location descriptions?
A: C9: Plano; C2: Custer Rd
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Sentence: A short video of the aftermath of HurricaneHarvey in Port Aransas on Cotter Ave. from Alister to Station St., 9/3/…
Q: Which parts of this sentence represent location descriptions?
A: C9: Port Aransas; C11: Cotter Ave. from Alister to Station St.
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Sentence: First St Baptist Church needs water & cleaning supplies for Port Arthur community flooded by Harvey
Q: Which parts of this sentence represent location descriptions?
A: C8: First St Baptist Church; C9: Port Arthur
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Sentence: Can anyone get food to memorial Hermann hospital at Gessner Rd in Houston? Request from CajunNavy hurricaneHarvey …
Q: Which parts of this sentence represent location descriptions?
A: C8: memorial Hermann hospital; C2: Gessner Rd; C9: Houston
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Sentence: Garth road & IH10 at Baytown, Tx. Very High water. Baytown REPORT ON harvey2017
Q: Which parts of this sentence represent location descriptions?
A: C5: Garth road & IH10; C9: Baytown; C9: Tx; C9: Baytown
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Sentence: someone placed a car here for ins purpose. parking lot of I-10 highway. Terry Hershey Park Parking lot. …
Q: Which parts of this sentence represent location descriptions?
A: C3: I-10 highway; C7: Terry Hershey Park
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Sentence: Need trailers/trucks to move dogs from Park Location: Whites Park Pavillion off I-10 exit 61 Anahuac TX
Q: Which parts of this sentence represent location descriptions?
A: C7: Whites Park Pavillion; C3: I-10; C4: exit 61; C9: Anahuac; C9: TX
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Sentence: 59 north, towards the Polk St exit: Go to Convention Center. houstonflood KHOU11 288 texasflood
Q: Which parts of this sentence represent location descriptions?
A: C3: 59 north; C4: Polk St exit; C7: Convention Center
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Sentence: Community is responding at shelters in College Park High School and Magnolia High School in TheWoodlands Harvey …
Q: Which parts of this sentence represent location descriptions?
A: C8: College Park High School; C8: Magnolia High School; C9: TheWoodlands'
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Sentence: FYI to any of you in NW Houston/Lakewood Forest, Projections are showing Cypress Creek overflowing at Grant Rd
Q: Which parts of this sentence represent location descriptions?
A: C10: NW Houston/Lakewood Forest; C5: Cypress Creek overflowing at Grant Rd
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Sentence: Father in law in a wheelchair. Send help to 6312 Bapling Drive, Sugar Land, TX HarveyStorm HarveyRelief HARVEYHELP HoustonStrande
Q: Which parts of this sentence represent location descriptions?
A: C1: 6312 Bapling Drive, Sugar Land, TX
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Sentence: Gas is apparently going up this week in San Antonio because of HurricaneHarvey, it's still 2.51 at the co-op on 17th & 36th ave
Q: Which parts of this sentence represent location descriptions?
A: C9: San Antonio; C5: 17th & 36th ave
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Sentence: White Oak Bayou around Stude Park. HoustonFloods Houston Heights
Q: Which parts of this sentence represent location descriptions?
A: C6: White Oak Bayou; C7: Stude Park; C9: Houston
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Sentence: Worried. A foot 2 go before Oyster Creek starts spilling into Lakes of Brightwater, Lakefront Drive, in front of our house …
Q: Which parts of this sentence represent location descriptions?
A: C6: Oyster Creek; C6: Lakes of Brightwater; C2: Lakefront Drive
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Sentence: Texas City asking 18 homes on S. Humble Camp Rd. to evacuate in case GCWA reservoir breaches Harvey
Q: Which parts of this sentence represent location descriptions?
A: C9: Texas City; C2: S. Humble Camp Rd.; C7: GCWA reservoir
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Sentence: Major flooding at Clay Rd & Queenston in west Houston. Lots of rescues going on for ppl trapped...
Q: Which parts of this sentence represent location descriptions?
A: C5: Clay Rd & Queenston; C9: Houston
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Sentence: Closed due to train derailment in Harvey on Destrehan Ave between River Rd and 4th St traffic NOLA
Q: Which parts of this sentence represent location descriptions?
A: C11: Destrehan Ave between River Rd and 4th St
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Sentence: Drop Off Location: The Life Center, Charlotte, 2435 Toomey Ave. #SocksForHouston LCFellowship …
Q: Which parts of this sentence represent location descriptions?
A: C7: The Life Center; C9: Charlotte; C1: 2435 Toomey Ave
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Sentence: Do we know how Minute Maid Park and NRG Park are?? Are they flooded as well as Honda Center? hurricaneharvey
Q: Which parts of this sentence represent location descriptions?
A: C7: Minute Maid Park; C7: NRG Park; C7: Honda Center
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Sentence: Eye wall still very much intact as Harvey is nearly stationary north of Rockport, TX. It's going to be a long nigh …
Q: Which parts of this sentence represent location descriptions?
A: C9: Rockport; C9: TX
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Sentence: {TEXT}
Q: Which parts of this sentence represent location descriptions?
A:

Table S3. Complete prompt created based on a set of 22 synthesized tweet examples using the same geo-knowledge.

This is a set of location description recognition problems.
The `Sentence` is a sentence containing location descriptions.
The goal is to infer which parts of the sentence represent location descriptions and the categories of the location descriptions. Split different location descriptions with `;`.
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Sentence: Please help family needs rescue at 112 Wikleson Dr, Houston 42143
Q: Which parts of this sentence represent location descriptions?
A: C1: 112 Wikleson Dr, Houston 42143
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Sentence: Heavy flooding in the Diego City/Amherst area. Please don't go there
Q: Which parts of this sentence represent location descriptions?
A: C10: Diego City/Amherst
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Sentence: Home Depot on Niagara Rd is donating tools for house repair #Harvey
Q: Which parts of this sentence represent location descriptions?
A: C2: Niagara Rd
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Sentence: There is an accident in Port Aransas on Bailey Rd. between Airport Dr and Station St
Q: Which parts of this sentence represent location descriptions?
A: C9: Port Aransas; C11: Bailey Rd. between Airport Dr and Station St
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Sentence: Water & cleaning supplies are needed at Grace Family Bible Church. Please help out...
Q: Which parts of this sentence represent location descriptions?
A: C8: Grace Family Bible Church
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Sentence: Help needed at memorial Hermann hospital at Gessner Rd in Houston! People need water and food there hurricaneHarvey …
Q: Which parts of this sentence represent location descriptions?
A: C8: memorial Hermann hospital; C2: Gessner Rd; C9: Houston

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Sentence: The intersection of Main road & I-60 is flooded. Very High water. harvey2017
Q: Which parts of this sentence represent location descriptions?
A: C5: Main road & I-60
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Sentence: A person is trapped at the parking lot off I-10 highway near Buffalo Bayou park …
Q: Which parts of this sentence represent location descriptions?
A: C3: I-10 highway; C7: Buffalo Bayou park
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Sentence: Exit 53 and exit 54 of I-11 are both flooded. Please avoid these two exits. Houston
Q: Which parts of this sentence represent location descriptions?
A: C4: Exit 53; C4: exit 54; C3: I-11; C9: Houston
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Sentence: 63 south, towards the Main St exit: Go to Convention Center. houstonflood 288 texasflood
Q: Which parts of this sentence represent location descriptions?
A: C3: 63 south; C4: Main St exit; C7: Convention Center
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Sentence: Shelters are provided in Williamsville High School. Please go there if you need a place to stay …
Q: Which parts of this sentence represent location descriptions?
A: C8: Williamsville High School
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Sentence: For those of you living in NW Houston/Lakewood Forest, Grant Rd @ Cypress Rd is flooded. Avoid it!
Q: Which parts of this sentence represent location descriptions?
A: C10: NW Houston/Lakewood Forest; C5: Grant Rd @ Cypress Rd
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Sentence: Grandpa needs help at 1831 West Ridge Rd, Sugar Land, TX HarveyStorm HarveyRelief HARVEYHELP HoustonStrande
Q: Which parts of this sentence represent location descriptions?
A: C1: 1831 West Ridge Rd, Sugar Land, TX
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Sentence: High water at the 11th & 30th ave. Houston is battered by heavy rain
Q: Which parts of this sentence represent location descriptions?
A: C5: 11th & 30th ave; C9: Houston
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Sentence: White Oak Bayou around Clear Lake Park. HoustonFloods Houston Heights
Q: Which parts of this sentence represent location descriptions?
A: C6: White Oak Bayou; C7: Clear Lake Park; C9: Houston
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Sentence: This is Buffalo Creek running under Kingsview Bridge near Frontier Drive, crazy …
Q: Which parts of this sentence represent location descriptions?
A: C6: Buffalo Creek; C7: Kingsview Bridge; C2: Frontier Drive
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Sentence: Families living on S. Washington Rd. will have to evacuate before the water gets too high #Harvey
Q: Which parts of this sentence represent location descriptions?
A: C2: S. Washington Rd.
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Sentence: Major flooding at High Rd & 15 Ave. in west Houston. People trapped. Please help...
Q: Which parts of this sentence represent location descriptions?
A: C5: High Rd & 15 Ave.; C9: Houston
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Sentence: Highway 10 between River Rd and 4th St is flooded. Please use local roads NOLA
Q: Which parts of this sentence represent location descriptions?
A: C11: Highway 10 between River Rd and 4th St
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Sentence: If you have things to donate, you can drop them off at ClearField Community Center, Houston, 351 7th Ave.
Q: Which parts of this sentence represent location descriptions?
A: C7: ClearField Community Center; C9: Houston; C1: 351 7th Ave
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Sentence: Hurricane eye wall is approaching Port Aransas, TX. Pray...
Q: Which parts of this sentence represent location descriptions?
A: C9: Port Aransas; C9: TX
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Sentence: Do we know the flooding conditions at Buffalo Bayou Park and Minute Maid Park? Any information can help.
Q: Which parts of this sentence represent location descriptions?
A: C7: Buffalo Bayou Park; C7: Minute Maid Park
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Sentence: {TEXT}
Q: Which parts of this sentence represent location descriptions?
A:

Table S4. Results of the Geo-GPT-4 model based on two additional sets of tweet examples.

Models	Precision	Recall	F-score
*Geo-GPT-4_set2**	0.701	0.737	0.718
Geo-GPT-4_syn	0.699	0.707	0.703

* Note: Because a different set of 22 tweets are used in the prompt for this model, the precision, recall, and F-score of *Geo-GPT-4_set2* are calculated based on a slightly different set of 978 tweets excluding those 22 tweets used in the prompt.

Ground Truth \ Predicted	C1	C2	C3	C4	C5	C6	C7	C8	C9	C10	C11	Missed
Door numbers: C1	240 91.25%	1 0.38%	0 0.00%	0 0.00%	0 0.00%	0 0.00%	0 0.00%	1 0.38%	2 0.76%	0 0.00%	0 0.00%	19 7.22%
Street names: C2	0 0.00%	145 64.73%	0 0.00%	0 0.00%	7 3.12%	0 0.00%	1 0.45%	0 0.00%	1 0.45%	0 0.00%	0 0.00%	70 31.25%
Highways: C3	0 0.00%	0 0.00%	13 50.00%	0 0.00%	1 3.85%	0 0.00%	0 0.00%	0 0.00%	0 0.00%	0 0.00%	0 0.00%	12 46.15%
Exits of highways: C4	0 0.00%	0 0.00%	0 0.00%	2 33.33%	0 0.00%	0 0.00%	0 0.00%	0 0.00%	0 0.00%	0 0.00%	0 0.00%	4 66.67%
Intersections: C5	0 0.00%	8 7.69%	1 0.96%	0 0.00%	71 68.27%	3 2.88%	0 0.00%	0 0.00%	0 0.00%	1 0.96%	1 0.96%	19 18.27%
Natural features: C6	0 0.00%	0 0.00%	0 0.00%	0 0.00%	0 0.00%	39 63.93%	1 1.64%	0 0.00%	3 4.92%	12 19.67%	0 0.00%	6 9.84%
Other human-made: C7	1 0.45%	12 5.38%	0 0.00%	0 0.00%	0 0.00%	3 1.35%	143 64.13%	10 4.48%	4 1.79%	21 9.42%	0 0.00%	29 13.00%
Local organizations: C8	0 0.00%	0 0.00%	0 0.00%	0 0.00%	0 0.00%	0 0.00%	1 1.59%	53 84.13%	0 0.00%	1 1.59%	0 0.00%	8 12.70%
Admin units: C9	0 0.00%	1 0.14%	0 0.00%	0 0.00%	3 0.42%	0 0.00%	2 0.28%	4 0.56%	463 65.21%	57 8.03%	0 0.00%	180 25.35%
Multiple areas: C10	0 0.00%	0 0.00%	0 0.00%	0 0.00%	0 0.00%	0 0.00%	0 0.00%	0 0.00%	0 0.00%	5 100.00%	0 0.00%	0 0.00%
Road segments: C11	0 0.00%	2 22.22%	0 0.00%	0 0.00%	3 33.33%	0 0.00%	0 0.00%	0 0.00%	0 0.00%	0 0.00%	1 11.11%	3 33.33%
Not in ground truth	26 6.70%	32 8.25%	9 2.32%	2 0.52%	38 9.79%	12 3.09%	47 12.11%	40 10.31%	123 31.70%	44 11.34%	4 1.03%	11 2.84%

Figure S1. Confusion matrix based on the output of Geo-GPT-4 and the annotated location descriptions using relaxed matching with 75% overlapping.